

Tutorial #5

Topics covered:

- The product and the box topologies
- The metric topology

Problem 1

Show that $(X_1 \times \cdots \times X_{n-1}) \times X_n$ is homeomorphic to $X_1 \times \cdots \times X_n$.

Problem 2

In the lectures we proved that given $f : A \rightarrow \prod_{\alpha \in J} X_\alpha$ by the equation

$$f(a) = (f_\alpha(a))_{\alpha \in J},$$

where $f_\alpha : A \rightarrow X_\alpha$ for each α and $\prod X_\alpha$ is endowed with the product topology, then the function f is continuous if and only if each function f_α is continuous. One of the two implications in this equivalence holds for the box topology. Which one? Prove it.

Problem 3

Let $\mathbb{R}^\infty$ be the subset of $\mathbb{R}^\omega$ consisting of all sequences that are “eventually zero”, that is, all sequences $(x_1, x_2, \dots)$ such that $x_i \neq 0$ for only finitely many values of i . What is the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ in the box and product topologies? Justify your answer.

Problem 4

Let X be metric space with metric d .

- (a) Show that $d : X \times X \rightarrow \mathbb{R}$ is continuous when $\mathbb{R}$ is endowed with the standard topology.
- (b) Let X' denote a topological space having the same underlying set as X but with a topology $\mathcal{T}'$ which is not necessarily the metric topology induced by d .

Show that if $d : X' \times X' \rightarrow \mathbb{R}$ is continuous, then the topology of $\mathcal{T}'$ is finer than the metric topology of X . One can summarize the result of this exercise as follows: If X has a metric d , then the topology induced by d is the coarsest topology relative to which the function d is continuous.

Problem 5

Given the points $\mathbf{x} = (x_\alpha)_{\alpha \in \mathbb{N}}$ and $\mathbf{y} = (y_\alpha)_{\alpha \in \mathbb{N}}$ of $\mathbb{R}^\omega$, let us define a metric $\bar{\rho}$ on $\mathbb{R}^\omega$ by the equation

$$\bar{\rho}(\mathbf{x}, \mathbf{y}) = \sup \{ \bar{d}(x_\alpha, y_\alpha) \mid \alpha \in \mathbb{N} \},$$

where $\bar{d}$ is the standard bounded metric on $\mathbb{R}$. It is easy to check that $\bar{\rho}$ is indeed a metric; it is called the **uniform metric** on $\mathbb{R}^\omega$, and the topology it induces is called the **uniform topology**. In Theorem 20.4 in the textbook it is proved that the uniform topology on $\mathbb{R}^\omega$ is finer than the product topology and coarser than the box topology; these three topologies are all different.

Let now $\mathbb{R}^\infty$ be the subset of $\mathbb{R}^\omega$ consisting of all sequences that are eventually zero (see Problem 3). What is the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ in the uniform topology? Justify your answer.

Problem 6

Let $A \subset X$. If d is a metric for the topology of X , show that $d|_{A \times A}$ is a metric for the subspace topology on A .